图像模型与大语言模型间的柔性连接

不想重新练LLM，也不想重新练ViT ⇒ 冻结ViT+LLM，
降成本

只训练一个连接器做为桥梁

问题换问题，用query token从图像提信息

BLIP-2: Bootstrapping Language-Image Pre-training with Frozen Image Encoders and Large Language Models

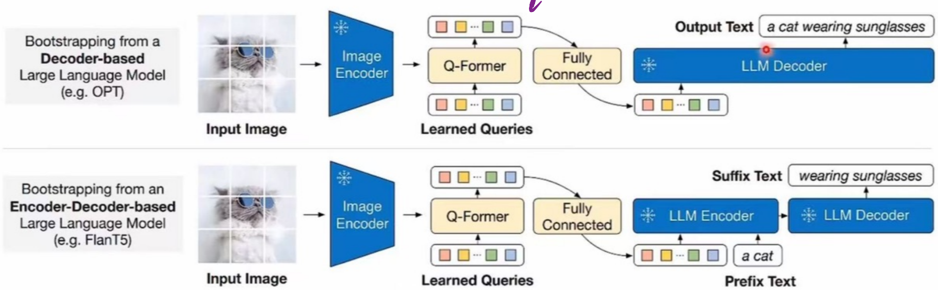
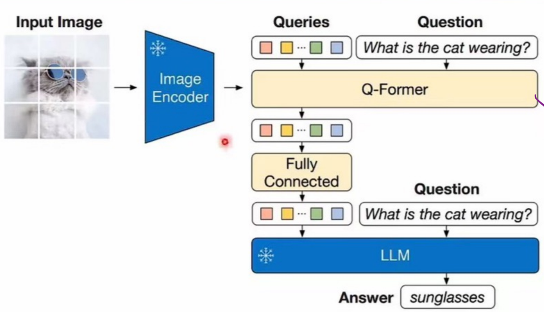


Figure 3. BLIP-2's second-stage vision-to-language generative pre-training, which bootstraps from frozen large language models (LLMs). (**Top**) Bootstrapping a decoder-based LLM (e.g. OPT). (**Bottom**) Bootstrapping an encoder-decoder-based LLM (e.g. FlanT5). The fully-connected layer adapts from the output dimension of the Q-Former to the input dimension of the chosen LLM.



不是输出答案

而是给LLM提供上下文

Figure 7. Model architecture for VQA finetuning, where the LLM receives Q-Former's output and the question as input, then predicts answers. We also provide the question as a condition to Q-Former, such that the extracted image features are more relevant to the question.

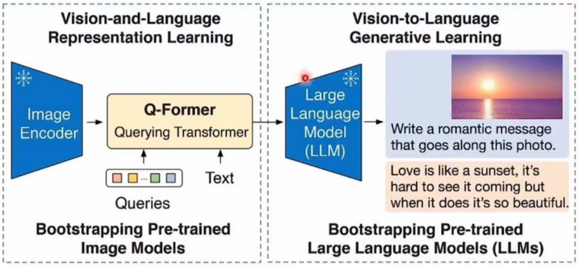


Figure 1. Overview of BLIP-2's framework. We pre-train a lightweight Querying Transformer following a two-stage strategy to bridge the modality gap. The first stage bootstraps vision-language representation learning from a frozen image encoder. The second stage bootstraps vision-to-language generative learning from a frozen LLM, which enables zero-shot instructed image-to-text generation (see Figure 4 for more examples).

让query学会提取对生成有用的信息。

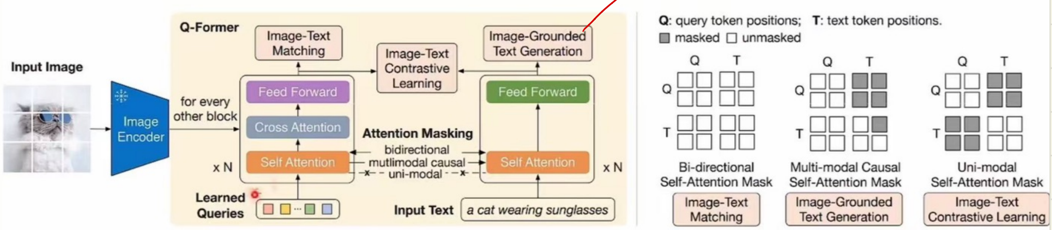


Figure 2. (**Left**) Model architecture of Q-Former and BLIP-2's first-stage vision-language representation learning objectives. We jointly optimize three objectives which enforce the queries (a set of learnable embeddings) to extract visual representation most relevant to the text. (**Right**) The self-attention masking strategy for each objective to control query-text interaction.